

Eric Yu

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Education

- 2025–2029 **Johns Hopkins University**, Baltimore, MD, USA
- **B.S.**, Molecular & Cellular Biology and Chemistry
 - **GPA:** 4.00/4.00
 - Dean's List: Fall 2025, Spring 2026
- 2023–2025 **Georgia Institute of Technology**, Atlanta, GA, USA
- Dual Enrollment in High School
 - **GPA:** 4.00/4.00

My relevant coursework can be found at the end of my CV [here](#).

Research Experience

- 2025– **Research Assistant**, [Huang Lab](#), Johns Hopkins University
- *Advisor:* Prof. Xiongyi Huang
 - Engineering proteins via directed evolution to access new-to-nature biocatalytic reactions, including azidation, CF₃ azidation, and skeletal editing reactions, enabling pharmaceutically important transformations
 - Utilizing computational tools (PyMOL, AutoDock Vina) to analyze protein structure and substrate binding interactions
 - Preparing substrate libraries through organic synthesis to evaluate the substrate scope of evolved enzymes
- 2024– **Research Assistant**, [Mao Lab](#), Indiana University School of Medicine
- *Advisor:* Prof. Weiming Mao
 - Designing a deep learning pipeline for automated segmentation and analysis of corneal endothelial cells and cross-linked actin networks with PyTorch
 - Assisting with literature review and the analysis of medical data, including those from optical coherence tomography (OCT), fundus imaging, and cell scratch assays

Other Experience

- 2025– **Violinist**, Hopkins Symphony Orchestra
- 2023–2025 **Violinist**, Atlanta Symphony Youth Orchestra

- Selected by a competitive multi-round audition process
 - Invited to perform at the 4th Annual Whitney E. Houston Foundation “Legacy of Love” Gala headlined by EGOT winner Jennifer Hudson
- 2022–2025 **Pianist & Violinist**, Georgia Philharmonic
- Performed Bernstein’s *Symphonic Dances from West Side Story* as a piano soloist
- 2021–2023 **Violinist**, Emory Youth Symphony Orchestra

Awards & Honors

- 2025 **National Merit Scholar**, \$2,500 Scholarship
- Selected from a pool of over 15,000 National Merit Finalists
- 2021–2025 **4-time National Medalist**, Science Olympiad National Tournament
- 8x Top 10 national tournament placements
 - 4x State champion team and national qualifier
 - Top 6, 9, and 14 national team placements as a leading member and captain
- 2024 **High Honors**, U.S. National Chemistry Olympiad (USNCO)
- Top 50 out of over 16,000 participants nationally (top ~0.3%) in the most prestigious national chemistry competition
 - #1 in the state of Georgia on the Local Exam
- 2024 **Qualifier/Invitee**, American Invitational Mathematics Exam (AIME)
- Top scorer on the American Mathematics Competition (AMC)
- 2024 **Gold Award**, The President’s Volunteer Service Award
- Awarded to those with 250+ volunteer hours in a 12 month period
 - Previously won a silver award in 2022
- 2024 **Prize Winner**, Georgia Music Teacher’s Association
- Award of Excellence in both the Instrumental Concerto and String competitions
- 2021–2023 **First Prize**, Sonatina & Sonata International Youth Piano Competition
- Prize winner of the XIV and XV editions
 - 2x Carnegie Hall performance opportunities as winner

Volunteering

- 2025– **Event Supervisor**, Science Olympiad
- Test writer and grader for Maryland state and regional tournaments, Johns Hopkins invitational

- 2025– **Musicare**
 - Music performance at local hospitals and nursing homes
- 2025– **MedShare & SHARE**
 - Sorting medical supplies for distribution to developing nations
- 2025– **Mural Art, Foundation for Hospital Art**
 - Painting murals to donate to hospitals and patients
- 2025– **Circle K International**
 - Volunteering to help local communities in Baltimore
- 2022–2025 **Coach and Test Writer, Science Olympiad**
 - Coached my middle school team in physics events, guiding them to win the state championship and a national silver medal
 - Volunteered to write and grade tests for top invitational tournaments across the country

Skills

Programming & Software

- Programming Languages: Python, MATLAB
- Machine Learning/AI: PyTorch, scikit-learn
- Scientific & Technical Software: PyMOL, ImageJ, AutoDock Vina, ChemDraw, Microsoft Office 365
- Typesetting: L^AT_EX

Laboratory-related

- Organic Synthesis: Column chromatography, TLC, inert atmosphere (Schlenk line), liquid-liquid extractions, recrystallization, rotary evaporation
- Biomolecular Techniques: Molecular cloning and library construction (PCR, mutagenesis, gel electrophoresis), bacterial culture & transformation (electroporation), protein expression & purification (sonication, affinity chromatography, SDS-PAGE), enzymatic assays
- Analytical Techniques: Chromatography (GC-MS, HPLC), spectroscopy (NMR, IR, UV-Vis, CD)

Publications

Detailed publication information can also be found at my [Google Scholar](#) **G**.

Journal Articles

- J1. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. Inhibition of pterygium cell fibrosis by the Rho kinase inhibitor. *Sci. Rep.* **14**, 30930 (2024). [DOI](#).

Peer-reviewed Conference Proceedings & Published Abstracts

- C1. Sugali, C. K., Dhamodaran, K., Dai, J., Harvey, D., Pushpala, N. M., **Yu, E.**, Huynh, K., Huang, A. Y. & Mao, W. Canonical Wnt signaling pathway inhibits glucocorticoid-induced ocular hypertension without compromising its anti-inflammatory function in the uveitis mouse model (2026).
– 2026 [ARVO](#) Annual Meeting in Denver, CO, May 3–7, 2026.
- C2. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. The Rho kinase inhibitor Y27632 inhibits unstimulated pterygium cell fibrosis. *Invest. Ophthalmol. Vis. Sci.* **66**, 6114 (2025). [LINK](#).
– 2025 [ARVO](#) Annual Meeting in Salt Lake City, UT, May 4–8, 2025.

Relevant Coursework

Johns Hopkins University

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| – Physical Chemistry I (CHEM 301) | – Genetics (BIOL 303) |
| – Physical Chemistry II (CHEM 302) | – Biochemistry I (BPHY 315) |
| – Intmd. Organic Chemistry Lab (CHEM 228) | – Protein Eng. & Biochemistry Lab (BPHY 253) |
| – Physical Chem. & Instrum. Lab (CHEM 307) | – Biological Physics (PHYS 310) |
| – Biological Models & Simulations (BMED 242) | – Contemporary Physics Seminar (PHYS 203) |

Georgia Institute of Technology *(dual enrolled in high school)*

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| – Cell & Molecular Biology (BIOS 3450) | – Linear Algebra (MATH 1554) |
| – Organic Chemistry I (CHEM 2311) | – Multivariable Calculus (MATH 2551) |
| – Organic Chemistry II (CHEM 2312) | – Differential Equations (MATH 2552) |
| – Synthesis Lab I (CHEM 2380) | – Princ. & Appl. of Eng. Materials (MSE 2001) |
| – Intro to Modern Physics (PHYS 2213) | – Intro to Computing (CS 1301) |

Last updated: June 3, 2026